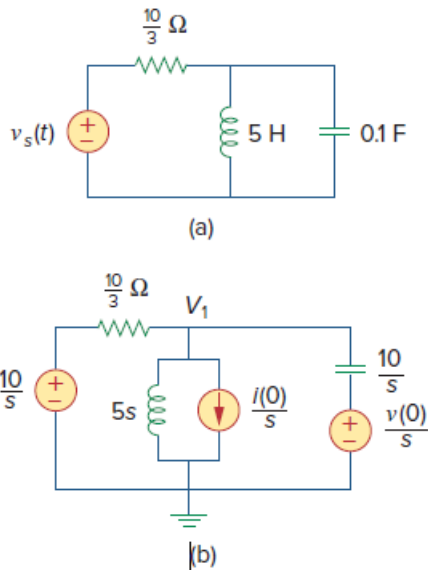


Problem

For the circuit shown in Fig. 16.12, and the same initial conditions in Example 16.4, find the current through the inductor for all time  $t > 0$  using superposition.

**Figure 16.12:**



**Example 16.4:**

Consider the circuit in Fig. 16.12(a). Find the value of the voltage across the capacitor assuming that the value of  $v_s(t) = 10u(t) \text{ V}$  and assume that at  $t = 0$ ,  $-1 \text{ A}$  flows through the inductor and  $+5 \text{ V}$  is across the capacitor

**Solution:**

Figure 16.12(b) represents the entire circuit in the  $s$ -domain with the initial conditions incorporated. We now have a straightforward nodal analysis problem. Because the value of  $V_1$  is also the value of the capacitor voltage in the time domain and is the only unknown node voltage, we only need to write one equation.

$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + \frac{i(0)}{s} + \frac{V_1 - [v(0)/s]}{1/(0.1s)} = 0 \quad (16.4.1)$$

or

$$0.1 \left( s + 3 + \frac{2}{s} \right) V_1 = \frac{3}{s} + \frac{1}{s} + 0.5 \quad (16.4.2)$$

where  $v(0) = 5 \text{ V}$  and  $i(0) = -1 \text{ A}$ . Simplifying we get

$$(s^2 + 3s + 2) V_1 = 40 + 5s$$

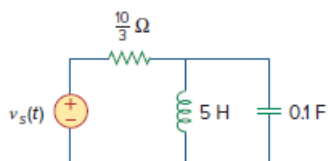
or

$$V_1 = \frac{40 + 5s}{(s + 1)(s + 2)} = \frac{35}{s + 1} - \frac{30}{s + 2} \quad (16.4.3)$$

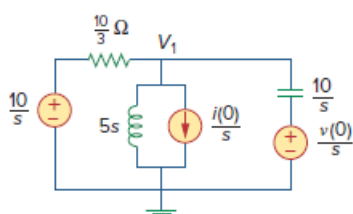
Taking the inverse Laplace transform yields

$$v_1(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V} \quad (16.4.4)$$

**Figure 16.12(a):**



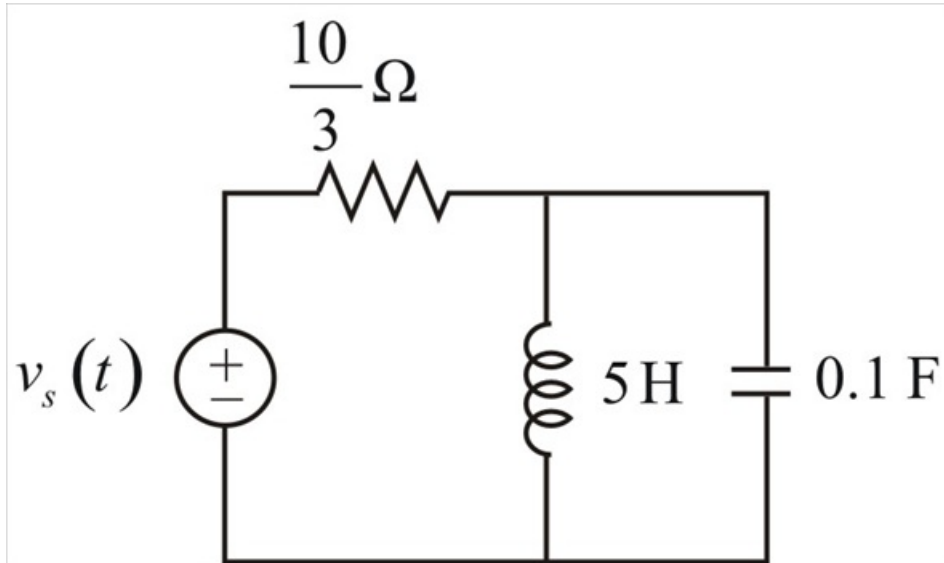
**Figure 16.12(b):**



## Step-by-step solution

### Step 1 of 13

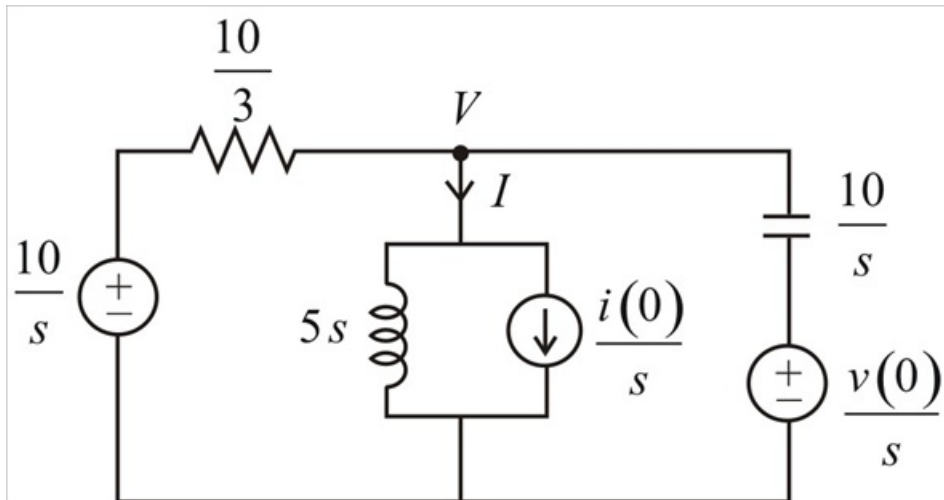
Consider the following given circuit:



### Step 2 of 13

The initial conditions are  $i(0) = -1 \text{ A}$  and  $v(0) = 5 \text{ V}$

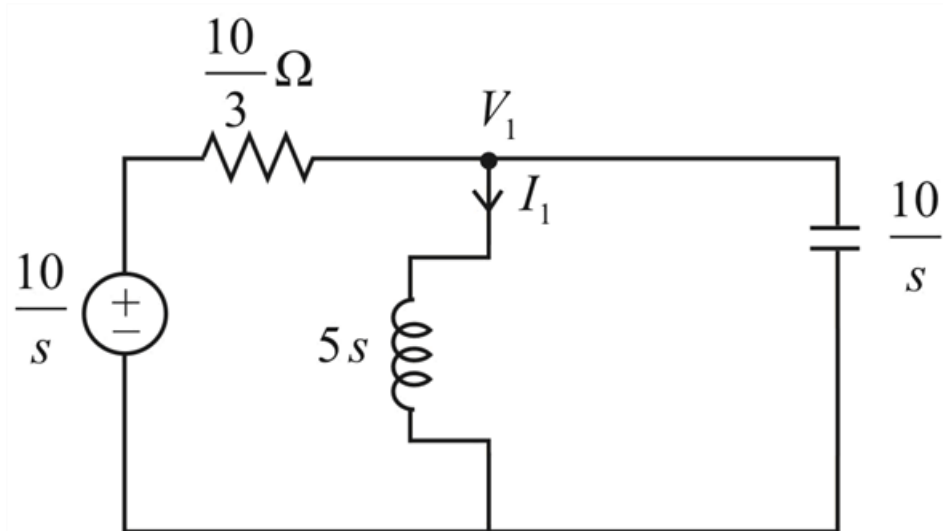
The following is the circuit in the s-domain with the initial conditions incorporated.



### Step 3 of 13

Use super position theorem, circuit considering one source at a time.

Consider  $\frac{10}{s}$  voltage source.



**Step 4** of 13

Apply Nodal analysis at the node.

$$\frac{V_1 - \frac{10}{s}}{\frac{10}{3}} + \frac{V_1}{5s} + \frac{V_1}{\frac{10}{s}} = 0$$

$$\frac{3V_1}{10} - \frac{30}{10s} + \frac{V_1}{5s} + \frac{sV_1}{10} = 0$$

$$V_1 \left( \frac{3}{10} + \frac{1}{5s} + \frac{s}{10} \right) = \frac{30}{10s}$$

$$V_1 \left( \frac{3s + 2 + s^2}{10s} \right) = \frac{30}{10s}$$

$$V_1 = \frac{30}{s^2 + 3s + 2}$$

$$V_1 = \frac{30}{(s+1)(s+2)}$$

**Step 5** of 13

Determine the current through the inductor for one source at a time.

$$I_1 = \frac{V_1}{5s}$$

$$I_1 = \left\{ \frac{30}{(s+1)(s+2)} \right\} \left\{ \frac{1}{5s} \right\}$$

$$I_1 = \frac{6}{s(s+1)(s+2)}$$

Apply partial fraction expansion.

$$\frac{6}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

Multiply  $s(s+1)(s+2)$  on both sides, we get

$$6 = A(s+1)(s+2) + B(s+2)s + Cs(s+1)$$

Substitute  $s = 0$  in the equation.

$$A(1)(2) = 6$$

$$A = 3$$

Substitute  $s = -1$

$$B(-1)(1) = 6$$

$$B = -6$$

Substitute  $s = -2$

$$(-2)(-1)C = 6$$

$$C = 3$$

#### Step 6 of 13

The current  $I_1$  is,

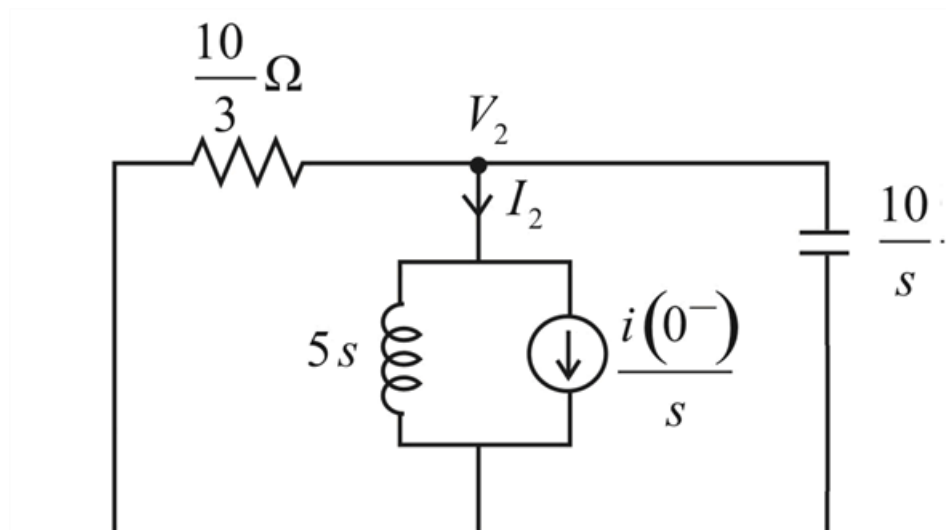
$$I_1 = \frac{3}{s} - \frac{6}{s+1} + \frac{3}{s+2}$$

Take inverse Laplace transform.

$$i_1(t) = (3 - 6e^{-t} + 3e^{-2t})u(t) \text{ A} \dots\dots (1)$$

#### Step 7 of 13

Consider the second source,  $\frac{i(0^-)}{s} = \frac{-1}{s}$



Apply Nodal analysis to the node.

$$\frac{V_2}{10} + \frac{V_2}{5s} + \frac{(-1)}{s} + \frac{V_2}{10} = 0$$

$$\frac{3V_2}{10} + \frac{V_2}{5s} + \frac{sV_2}{10} = \frac{1}{s}$$

$$V_2 \left[ \frac{3s+2+s^2}{10s} \right] = \frac{1}{s}$$

$$V_2 = \frac{10}{(s+1)(s+2)}$$

## Step 9 of 13

Determine the current through the inductor with the second source.

$$I_2 = \frac{V_2}{5s} + \frac{i(0^-)}{s}$$

$$I_2 = \left[ \frac{10}{(s+1)(s+2)} \right] \left[ \frac{1}{5s} \right] - \frac{1}{s}$$

$$I_2 = \frac{2}{s(s+1)(s+2)} - \frac{1}{s}$$

Use partial fraction expansion.

$$\frac{2}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$2 = A(s+1)(s+2) + Bs(s+2) + Cs(s+1)$$

Substitute  $s = 0$  in the equation.

$$(1)(2)A = 2$$

$$A = 1$$

Substitute  $s = -1$

$$(-1)(1)B = 2$$

$$B = -2$$

Substitute  $s = -2$

$$(-2)(-1)C = 2$$

$$C = 1$$

## Step 10 of 13

The current  $I_2$  is,

$$I_2 = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2} - \frac{1}{s}$$

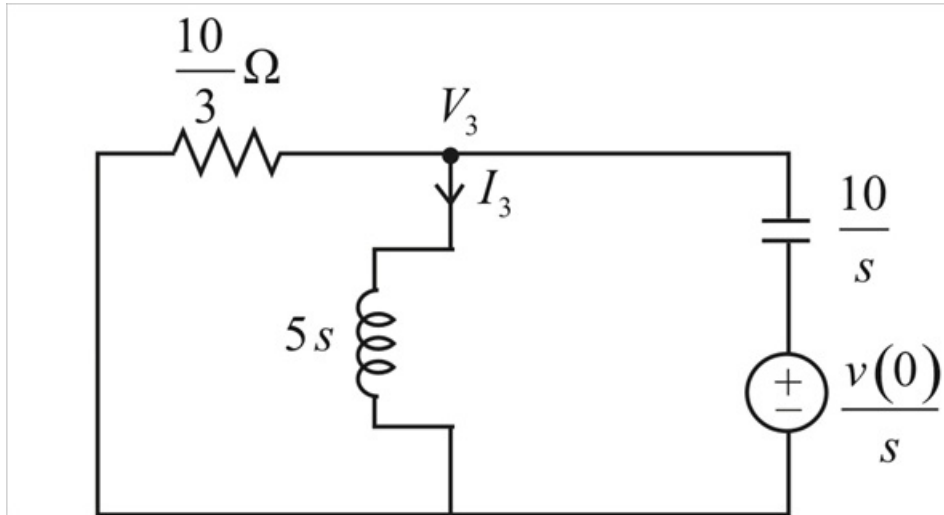
$$= \frac{-2}{s+1} + \frac{1}{s+2}$$

Take inverse laplace transform.

$$i_2(t) = (e^{-2t} - 2e^{-t})u(t) \text{ A} \dots\dots (2)$$

## Step 11 of 13

Consider the third source  $\frac{v(0)}{s} = \frac{5}{s}$



Apply Nodal analysis to the top node.

$$\frac{V_3}{\frac{10}{3}} + \frac{V_3}{5s} + \frac{V_3 - \frac{5}{s}}{\frac{10}{s}} = 0$$

$$\frac{3V_3}{10} + \frac{V_3}{5s} + \frac{sV_3}{10} - \frac{5}{10} = 0$$

$$V_3 \left( \frac{3s + 2 + s^2}{10s} \right) = \frac{1}{2}$$

$$V_3 = \frac{5s}{(s+1)(s+2)}$$

Thus, the current through the inductor for the third source is,

$$\begin{aligned} I_3 &= \frac{V_3}{5s} \\ &= \left[ \frac{5s}{(s+1)(s+2)} \right] \left[ \frac{1}{5s} \right] \\ &= \frac{1}{(s+1)(s+2)} \end{aligned}$$

Use partial fraction expansion.

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$1 = A(s+2) + B(s+1)$$

Substitute  $s = -1$  in the equation.

$$A = 1$$

Substitute  $s = -2$

$$(-1)B = 1$$

$$B = -1$$

The current  $I_3$  is,

$$I_3 = \frac{1}{s+1} - \frac{1}{s+2}$$

Take inverse Laplace transform

$$i_3(t) = (e^{-t} - e^{-2t})u(t) \text{ A} \dots\dots (3)$$

### Step 13 of 13

Add all the three equations are (1), (2) and (3).

$$\begin{aligned} i(t) &= i_1(t) + i_2(t) + i_3(t) \\ &= (3 - 6e^{-t} + 3e^{-2t} + e^{-2t} - 2e^{-t} + e^{-t} - e^{-2t})u(t) \\ &= 3u(t) + \{(-6 - 2 + 1)e^{-t} + (3 + 1 - 1)e^{-2t}\}u(t) \\ &= (3 - 7e^{-t} + 3e^{-2t})u(t) \text{ A} \end{aligned}$$

Thus, the current through the inductor for all time  $t > 0$  is,

$$\boxed{i(t) = (3 - 7e^{-t} + 3e^{-2t})u(t) \text{ A}}.$$